



FORT STREET HIGH SCHOOL

2006

HIGHER SCHOOL CERTIFICATE COURSE

ASSESSMENT TASK 4: TRIAL HSC

Mathematics

Time allowed: 3 hours

(Plus 5 minutes reading time)

Outcomes Assessed	Questions	Marks
Chooses and applies appropriate mathematical techniques in order to solve problems effectively	1, 2	
Manipulates algebraic expressions to solve problems from topic areas such as functions, quadratics, probability and series	4, 6, 8	
Demonstrates skills in the processes of differential and integral calculus and applies them appropriately	3, 5, 7	
Synthesises mathematical solutions to harder problems and communicates them in appropriate form	9, 10	

Question	1	2	3	4	5	6	7	8	9	10	Total	%
Marks	/12	/12	/12	/12	/12	/12	/12	/12	/12	/12	/120	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each new question is to be started on a new page

Name: _____

Teacher: _____

Class: _____

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1 (12 marks) (Start a new page)

a) Given that $v = \frac{1}{\sqrt{t^2 + 1}}$ find the value of v when $t = 2.04 \times 10^3$, giving your answer in scientific notation correct to 2 significant figures. 2

b) Find those values of x which satisfy the inequality $|5 - 2x| \geq 3$. 2

c) Given $\log_6 2 = 0.3869$ and $\log_6 3 = 0.6131$ find

i) $\log_6(1.5)$ 1

ii) $\log_6 24$ 1

d) Simplify: $\frac{a^2 - a}{a} \times \frac{a^2 + 2a + 1}{a^2 - 1}$ 2

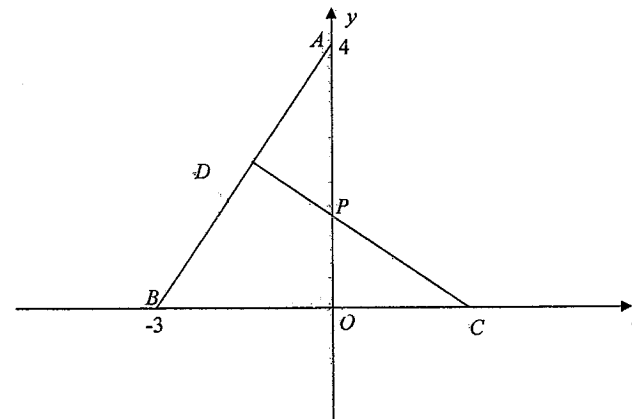
e) If $f(x) = \begin{cases} -2 & x \leq -3 \\ x+1 & -3 < x < 1 \\ x^2 & x \geq 1 \end{cases}$

i) Find the value of $f(-3) + f(0) - f(2)$ 1

ii) Write down an expression for $f(a^2 + 1)$ 1

f) Find x if $\log_{16} 2 = \log_x 3$ 2

Question 2: (12 marks) (Start a new page)



In the diagram $AB = BC$ and CD is perpendicular to AB .

CD intersects the y axis at P .

Copy the diagram onto your answer sheet.

a) Find the length of AB 1

b) Hence show the co-ordinates of C are $(2, 0)$ 1

c) Show the equation of CD is $3x + 4y = 6$. 3

d) Show that the co-ordinates of P are $(0, 1\frac{1}{2})$. 1

e) Use Pythagoras' theorem on $\triangle POC$ to show the length of CP is $2\frac{1}{2}$ units. 1

f) Prove $\triangle ADP$ is congruent to $\triangle COP$. 3

g) Hence calculate the area of the quadrilateral $DPOB$. 2

Question 3: (12 marks) (Start a new page)

a) Differentiate

i) $y = x\sqrt{x}$

ii) $y = \frac{e^x + 2x}{x}$

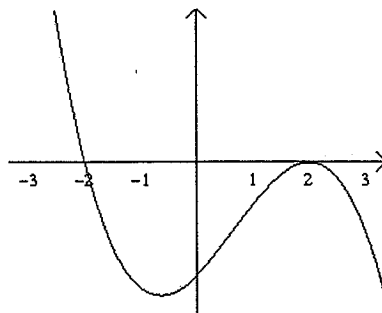
iii) $y = (3 - 4x)^5$

b) Find

i) $\int e^{\frac{x}{2}} dx$

ii) $\int_1^2 \left(2x - \frac{1}{x^2} \right) dx$

c) The figure below shows the graph of $y = f'(x)$ where $f'(x)$ is the derivative of a function $f(x)$. The domain of $f(x)$ and $f'(x)$ is $-3 \leq x \leq 3$



Copy the diagram onto your answer sheet.

i) By considering the graph of $y = f'(x)$, explain why the graph of $y = f(x)$ has two, and only two stationary points.

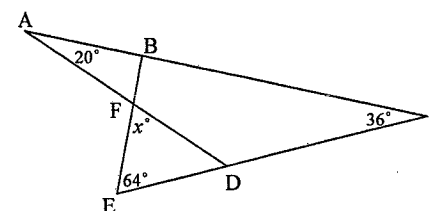
ii) For what values of x does $y = f(x)$ have a relative maximum? Justify your answer.

iii) Given that $f(-3) = 0$ sketch a possible graph of $y = f(x)$ on the same axes that you drew the graph $y = f'(x)$.

Question 4: (12 marks) (Start a new page)

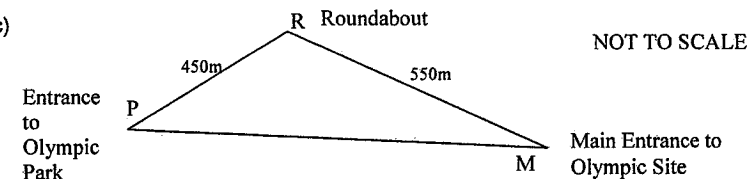
a) Evaluate $\sum_{n=5}^{14} 17 - 2n$.

b)



Find the value of x , giving reasons.

c)



The diagram shows the route MPR, followed by a shuttle bus from the main entrance to the Sydney Olympic Site (M), via a roundabout (R), to the entrance to Olympic Park (P), where the principal stadiums are sited. R is 550 metres at a bearing of 325° from M, and P is 450 metres at a bearing of 250° from R. It is proposed that an overhead cable car be built directly from M to P.

i) Copy the diagram onto your page and show that $\angle MRP$ measures 105° .

ii) Calculate the distance MP, covered by the cable car (to the nearest m).

d) Let α and β be the roots of the equation $x^2 - 5x + 2 = 0$. Find the values of

i) $\alpha + \beta$

ii) $\alpha\beta$

iii) $(\alpha + 1)(\beta + 1)$

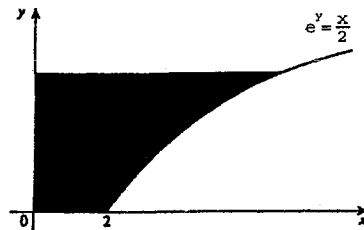
e) Solve $2\sin\theta = -1$ for $0^\circ \leq \theta^\circ \leq 360^\circ$

Question 5: (12 marks) (Start a new page)

- a) The curve $y = ax^3 + \frac{b}{x^2}$ cuts the x-axis at $x = 1$, and the gradient of the tangent at this point is 2. Find the values of a and b .

3

b)

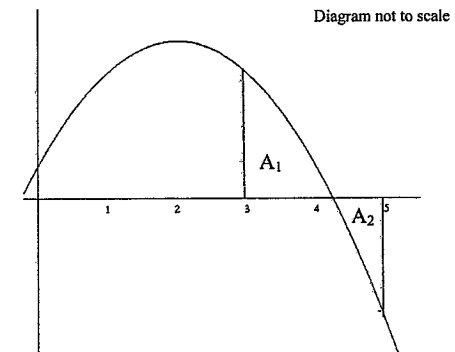


The shaded region shown in the diagram is bounded by the x and y axes, the curve $e^y = \frac{x}{2}$ and the line $y = 1$.

A solid is formed by rotating the shaded region about the y axis. Show that the volume of the solid is given by $4\pi \int_0^1 e^{2y} dy$ and find the volume correct to 2 decimal places.

3

- c) The graph of $y = 1 + 4x - x^2$ over the domain $0 \leq x \leq 5$ is shown below



- i) What is the range of y over this domain?

1

- ii) Evaluate $\int_3^5 (1 + 4x - x^2) dx$ and hence determine which area A_1 or A_2 is greater? Give reasons.

3

- d) Show that $\frac{\sec^2 \theta}{\tan^2 \theta} = \frac{1}{\sin^2 \theta}$.

2

Question 6: (12 marks) (Start a new page)

- a) For what values of m does the quadratic equation

$$(5m-3)x^2 - 4mx + m + 1 = 0$$

have one real root.

3

- b) Find the coordinates of the vertex and the focus and the equation of the directrix of the parabola $y^2 - 2y - 15 = 4x$.

3

- c) A caterer organises parties for groups of up to 200. She calculates the cost price of a party by allowing \$22 per head for the first 10 guests, \$21 per head for the next 10 guests, and so on, allowing one dollar less per head for each subsequent group of 10 guests or part thereof.

- i) Show that the cost price, in dollars, for each guest in the n^{th} group of 10 guest, or part thereof, is given by

$$T_n = 23 - n$$

where T_n is the n^{th} term of an arithmetic series.

1

- ii) Find the increase in the cost price of a party if 4 more persons are added to a guest list of 85.

1

- iii) Determine the cost price of a party attended by 115 people.

2

- iv) If the caterer wishes to make a 25% profit on the cost price, calculate the average charge per head (to the nearest 5 cents) for a party of 115 guests.

2

Question 7: (12 marks) (Start a new page)

- a) A defence system of a city has both ground-to-air missiles and anti-aircraft guns. The missiles have a 0.85 chance of hitting an attacking plane while the anti-aircraft guns have a 0.2 chance of hitting it.

- i) A single plane attacks the city. Find the probability that it will not be shot down by either the missile or the gun.

2

- ii) If two planes attack the city, what is the probability at least one is shot down?

2

- b)

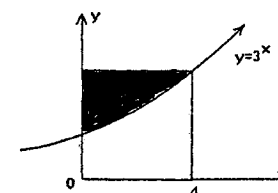
- i) Use the Trapezoidal Rule with 2 strips (i.e. 3 function value), to find an

approximate value for $\int_0^4 3^x dx$.

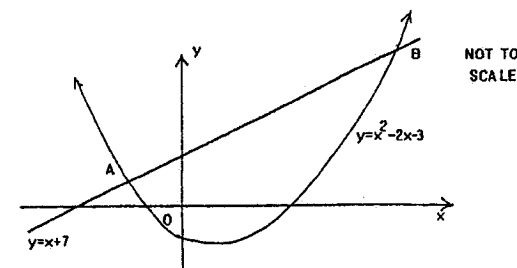
2

- ii) Hence find an approximate value for the shaded area in the diagram below.

2



- c) The diagram shows the graphs $y = x^2 - 2x - 3$ and $y = x + 7$. The graphs intersect at the points A and B.



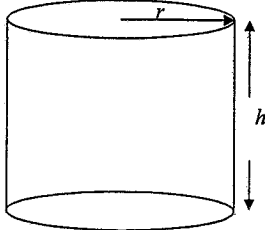
- i) Find the coordinates of A and B.

2

- ii) Find the area enclosed by $y = x^2 - 2x - 3$ and $y = x + 7$.

2

Question 8: (12 marks) (start a new page)

- a) A PIN number for a security code consists of a 3 digit number where each number can be any of the numbers (0, 1, 2, 3, 4, 5, 6, 7, 8, 9)
- i) Find the probability that a PIN number entered at random will break the code. 1
- ii) A thief knows that the middle number is 4 or 6. What is the probability that she will break the code (entering one PIN number at random)? 2
- b) A geometric series has a second term of 6 and the ratio of the seventh term to the sixth is 3.
- i) Find the common ratio r . 1
- ii) What is the first term a ? 1
- iii) Calculate the sum of the first 12 terms. 1
- c) With the breaking of the drought in the Narrabri area, Farmer Jo wants to install a new water tank. It is to be in the shape of a closed cylinder with radius r metres and height h metres as shown in the diagram. The surface area of metal to be used in the tank's construction is 30 square metres.
- 
- i) Find an expression for h in terms of r . 2
- ii) Show that the volume $V \text{ m}^3$ of the tank is given by the formula $V = 15r - \pi r^3$. 1
- iii) Find the radius (to the nearest cm) if the volume of the tank is to be maximised. 3

Question 9: (12 marks) (Start a new page)

- a) Don planted a Grevillia hedge. The plants were 20cm tall when he planted them. After 1 year they were 1m tall. The next year they grew 60% of the previous years growth to a height of 148cm. They continued to grow 60% of the previous year's growth each year until they reached their maximum height.
- i) What was the height of each Grevillia after 3 years? 1
- ii) Calculate the maximum height of the Grevillia hedge. 2
- b) Consider the curve given by the function $f(x) = x^2 + \frac{1}{x^2}$.
- i) State the domain of the function. 1
- ii) Determine if the function is odd or even.
- iii) Describe the behaviour of the curve as x gets very close to zero. 1
- iv) Find the coordinates of the stationary points and determine their nature. 4
- v) Explain why the curve is concave up for all values of x in its domain. 1
- vi) Sketch the curve for $-2 \leq x \leq 2$. 2

Question 10: (12 marks) (Start a new page)

a) The population of a certain town at the beginning of the year 2000 was 100 000. The population increases (due to births and new arrivals) by 12% each year, but also decreases (due to departures and deaths) by E people each year.

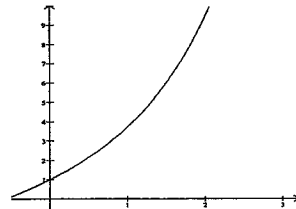
- i) Prove that the population at the beginning of 2001 was $112000 - E$. 1
 ii) Show that the population after 2 years is $125440 - 2.12E$. 1
 iii) If the population at the beginning of 2006 is 140000, calculate the decrease E due to departures and deaths each year (to the nearest whole person). 4

b)

i) Show that $\frac{d(xe^x)}{dx} = e^x + xe^x$. 1

ii) Hence, by integrating both sides of the equation in (i), show that $\int_0^2 xe^x dx = e^2 + 1$. 2

iii) The region that lies between the x -axis and the curve $y = e^x + x$ as shown in the diagram is rotated about the x -axis between $x = 0$ and $x = 2$ to form a solid.



Find the volume of this solid (leave your answer in exact form). 3

END OF EXAMINATION

Question 1.

$$a) V = \frac{1}{\sqrt{t^2 + 1}}$$

$$= \frac{1}{\sqrt{(2.04 \times 10^3)^2 + 1}}$$

$$= 2.40292 \times 10^{-7}$$

$$= 2.4 \times 10^{-7} \text{ (2 s.f.)}$$

$$b) |5 - 2x| \geq 3$$

$$5 - 2x \geq 3 \quad , \quad 5 - 2x \leq -3$$

$$-2x \geq -2 \quad , \quad -2x \leq -8$$

$$x \leq 1 \quad \checkmark \quad x \geq 4 \quad \checkmark$$

$$c) i) \log_6(15) = \log_6\left(\frac{3}{2}\right)$$

$$= \log_6 3 - \log_6 2$$

$$= 0.6131 - 0.3869$$

$$= 0.2262$$

$$ii) \log_6 24 = \log_6(6 \times 4)$$

$$= \log_6 6 + \log_6 4$$

$$= 1 + 2\log_6 2$$

$$= 1 + 2 \times 0.3869$$

$$= 1.7738$$

$$\text{OR } \log_6 24 = \log_6 3 + \log_6 8$$

$$= \log_6 3 + \log_6 2^3$$

$$= 0.6131 + 3 \times 0.3869$$

$$= 1.7738$$

$$d) \frac{a^2 - a}{a} \times \frac{a^2 + 2a + 1}{a^2 - 1} = \frac{a(a-1)}{a} \times \frac{(a+1)^2}{(a-1)(a+1)}$$

$$= a+1 \quad \checkmark$$

Q1 e) i) $f(3) + f(1) - f(2)$
 $= -2 + 1 - 4$
 $= -5$ ✓

ii) $f(a^2+1) = (a^2+1)^2$
 $= a^4 + 2a^2 + 1$ ✓

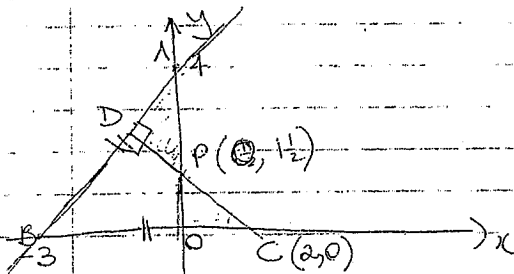
f) $\log_{16} 2 = \log_x 3$
 $\frac{1}{4} = \log_x 3$

$x^{\frac{1}{4}} = 3$ ✓

$x = 3^4$

$x = 81$ ✓

QUESTION 2



a) $|AB| = 5$ units (by Pythagoras in $\triangle ABO$) ✓

b) $BC = 5$ units ($AB = BC$)
 since B is $(-3, 0)$ then $-3 + 5 = 2$ ✓
 $C(2, 0)$

c) $A(0, 4)$ $B(-3, 0)$ $C(2, 0)$
 $m_{AB} = \frac{-4}{3}$
 $= -\frac{4}{3}$
 $\therefore m_{\perp} = -\frac{3}{4}$ (eqn $3x + 4y = 6$ (as req'd)) ✓

d) $3x + 4y = 6$
 cuts y axis $\Rightarrow x = 0$

$4y = 6$
 $y = \frac{3}{2} \Rightarrow P(0, \frac{3}{2})$ ✓

e) $CP^2 = 1.5^2 + 2^2$
 $= 2.25 + 4$
 $= 6.25$
 $CP = 2\frac{1}{2}$ units ✓

f) In $\triangle ADP$ & $\triangle COP$

i) $\angle ADP = \angle COP = 90^\circ$ (given) ✓

ii) $\angle DPA = \angle PCO$ (vert. opp.) ✓

iii) $AP = PC = 2\frac{1}{2}$ (proved above) ✓

$\therefore \triangle ADP \equiv \triangle COP$ (AAS) ✓

g) area quad DPOB = $\triangle APO - \triangle ADP - \triangle PCO$
 $= \frac{1}{2} \times 3 \times 4 - \frac{1}{2} \times 2 \times \frac{3}{2}$
 $= 6 - 1\frac{1}{2}$
 $= 4\frac{1}{2}$ u² ✓

QUESTION 3

a) i) $y = x^{\frac{3}{2}}$

$= x^{\frac{3}{2}}$

$\frac{dy}{dx} = \frac{3x^{\frac{1}{2}}}{2}$

$= \frac{3\sqrt{x}}{2}$

✓ (either form)

ii) $y = \frac{e^x + 2x}{x}$

$\frac{dy}{dx} = \frac{v \cdot \frac{du}{dx} - u \cdot \frac{dv}{dx}}{v^2}$

$= \frac{x(e^x + 2) - (e^x + 2x)}{x^2}$ ✓

$$\frac{dy}{dx} = \frac{xe^x + 2x - e^x - 2x}{x^2}$$

$$= \frac{xe^x - e^x}{x^2}$$

$$= \frac{e^x(x-1)}{x^2}$$

✓ (either form)

$$\text{iii)} \quad y = (3-4x)^5$$

$$\frac{dy}{dx} = 5(3-4x)^4 \cdot (-4)$$

$$= -20(3-4x)^4$$

✓

$$\text{b) i)} \quad \int e^{\frac{x}{2}} dx$$

$$= \frac{e^{\frac{x}{2}}}{\frac{1}{2}} + C$$

$$= 2e^{\frac{x}{2}} + C$$

✓

$$\text{ii)} \quad \int_1^2 \left(2x - \frac{1}{x^2} \right) dx$$

$$= \left[2x - x^{-2} \right]_1^2$$

$$= \left[x^2 + x^{-1} \right]_1^2$$

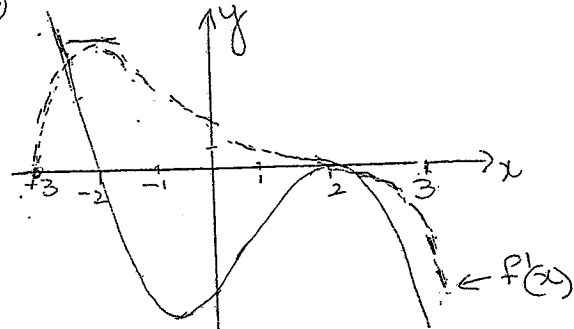
$$= \left[x^2 + \frac{1}{x} \right]_1^2$$

$$= \left(4 + \frac{1}{2} \right) - \left(1 + 1 \right)$$

$$= 2\frac{1}{2}$$

✓

Q3
c)



i) $y = f(x)$ has only 2 statnary pts at $x = -2$ and $x = 2$ as $f'(x)$ only cuts the x axis twice. (i.e. $f'(x) = 0$ twice) ✓

ii) $x = -2$. ✓

x	-2^-	-2	-2^+
$f'(x)$	+	0	-
$f(x)$	> 0	$f(x) = 0$	< 0

∴ a max turning pt. at $x = -2$. ✓

iii) $f(-3) = 0$

∴ $x = -3, y = 0$.

$$\underline{x=2}$$

x	2^-	2	2^+
$f(x)$	-	0	-

✓ +1 for graph

or 2 for showing features

Q4. a) $\sum_{n=5}^{14} 17 - 2n$

$= 7 + 5 + 3 + \dots + -11$

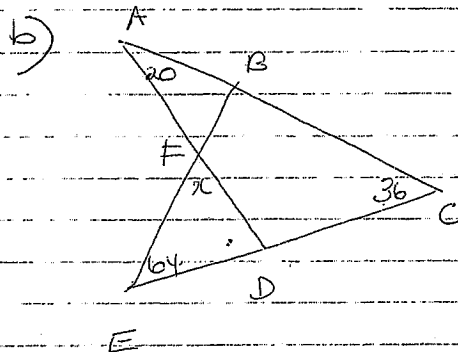
$a = 7, d = -2, n = 10$

$S_n = \frac{n}{2}(a + d)$

$S_{10} = \frac{10}{2}(7 - 11)$

$= 5 \times -4$

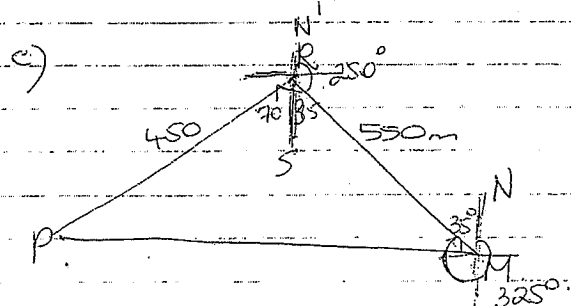
$= -20$



$\angle ADE = 56$ (ext. $\angle \triangle ACD$)

$x + 64 + 56 = 180$ ($\angle$ sum $\triangle EFD$)

$x = 60$



$\angle SRP = 250 - 180$
 $= 70^\circ$

$\angle SRM = 35^\circ$ (alt. $\angle$ s on $\parallel$ lines)

$\angle MRP = 35 + 70$

Q4

d) $MP^2 = 450^2 + 550^2 - 2 \times 450 \times 550 \times \cos 105^\circ$
 $= 63315.4273$

$MP = 795.6855$

$MP = 796 \text{ m}$ (nearest m)

d) $x^2 - 5x + 2 = 0$

$a = 1, b = -5, c = 2$

i) $x + b = \frac{-b}{a}$
 $= 5$

ii) $x + b = \frac{c}{a}$
 $= 2$

iii) $(x+1)(y+1) = xy + x + y + 1$
 $= 2 + 5 + 1$
 $= 8$

e) $2 \sin \theta = -1$ $0^\circ \leq \theta^\circ \leq 360^\circ$
 $\sin \theta = -\frac{1}{2}$
acute $\theta = 30^\circ$
 $\theta = 210^\circ, 330^\circ$

QUESTIONS

$$a) y = ax^3 + \frac{b}{x^2} \quad \text{cuts } x\text{-axis} \Rightarrow y = 0 \Rightarrow (1, 0)$$

$$\text{and } f'(1) = 2$$

$$0 = a + b \quad (1)$$

$$\frac{dy}{dx} = 3ax^2 - 2bx^{-3}$$

$$= 3ax^2 - \frac{2b}{x^3}$$

$$\text{when } x=1, \frac{dy}{dx} = 2$$

$$2 = 3a - 2b \quad (2)$$

$$\text{From (1) } a = -b$$

$$2 = -3b - 2b$$

$$-5b = 2$$

$$b = -\frac{2}{5}$$

$$\therefore a = \frac{2}{5}$$

$$b) e^y = \frac{x}{2}$$

$$x = 2e^y$$

$$x^2 = 4e^{2y}$$

$$V = \pi \int x^2 dy$$

$$V = \pi \int_0^1 4e^{2y} dy$$

$$V = \frac{4\pi}{2} [e^{2y}]_0^1$$

$$= 2\pi [e^2 - 1]$$

$$V = 40.14 \text{ u}^3 \quad (2 \text{ dp})$$

Q5

$$i) y = 1 + 4x - x^2$$

$$x=2 \quad y = 1 + 8 - 4 = 5$$

$$x=5 \quad y = 1 + 20 - 25 = -4$$

$$R: -4 \leq y \leq 5$$

$$ii) \int_3^5 (1 + 4x - x^2) dx$$

$$= \left[x + 2x^2 - \frac{x^3}{3} \right]_3^5$$

$$= (5 + 50 - \frac{125}{3}) - (3 + 18 - \frac{27}{3})$$

$$\therefore A_1 \text{ is greater than } A_2$$

as the overall area is positive

$$a) \frac{\sec^2 \theta}{\tan^2 \theta} = \frac{1}{\sin^2 \theta}$$

$$\text{L.H.S.} = \frac{1}{\cos^2 \theta} \div \tan^2 \theta$$

$$= \frac{1}{\cos^2 \theta} \div \frac{\sin^2 \theta}{\cos^2 \theta}$$

$$= \frac{1}{\cos^2 \theta} \times \frac{\cos^2 \theta}{\sin^2 \theta}$$

$$= \frac{1}{\sin^2 \theta}$$

QUESTION 6.

a) $(5m-3)x^2 - 4mx + m+1 = 0$

$a = 5m-3$ $b = -4m$ $c = m+1$

for one equal root $\Rightarrow \Delta = 0$

$b^2 - 4ac = 0$

$(-4m)^2 - 4(5m-3)(m+1) = 0$ ✓

$16m^2 - 4(5m^2 + 2m - 3) = 0$

$16m^2 - 20m^2 - 8m + 12 = 0$

$-4m^2 - 8m + 12 = 0$

$-(m^2 + 2m - 3) = 0$ ✓

$(m-1)(m+3) = 0$

$\therefore m = 1, m = -3$ for one real root ✓

b) $y^2 - 2y - 15 = 4x$

$y^2 - 2y + 1 = 4x + 15 + 1$

$(y-1)^2 = 4x + 16$

$(y-1)^2 = 4(x+4)$

cf $(y-k)^2 = 4(x+h)$ ✓

vertex $(-4, 1)$ ✓

$a = 1$

$F(-3, 1)$ ✓

directrix $x = -5$ ✓

c) $a = 22, d = 1$ $T_n = n^{\text{th}} \text{ multiple of } 10$

$T_n = a + (n-1)d$

$= 22 + (n-1) \times 1$

$= 22 - n + 1$

$T_n = 23 - n$ ✓

$T_9 = 23 - 9$

$= 14$

Extra cost = $4 \times \$14$

$= \$56$ ✓

Q6 iii) 111th person $\rightarrow$ 120th person

$= 12^{\text{th}}$ group of 10

$T_{12} = 23 - 12$

$= \$11/\text{head}$

101st person to 110th person

$= 11^{\text{th}}$ grp of 10

$T_{11} = 23 - 11$

$= \$12/\text{head}$

Cost = $10 \times [\underbrace{\$22 + \$21 + \$20 + \dots + \$12}_{+ \$11 \times 5}]$ OR fixed cost for 12 whole groups = 5 people

$= 10 \times \frac{n}{2}(a+l) + 55$

$= \frac{12}{2}(220 + 11 \times 10)$

$= \$1925$

$= 10 \times \frac{11}{2}(22+12) + 55$

$= 55 \times 34 + 55$ ✓

$= \$1925$

N) Cost + 25% profit = $1.25 \times \$1925$
 $= \$2406.25$ ✓

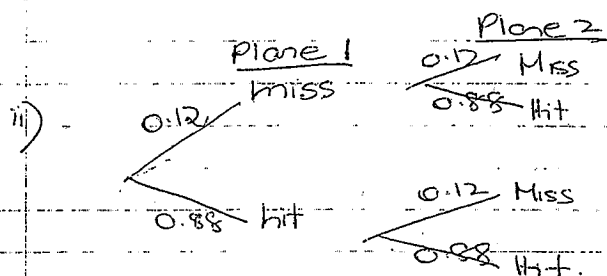
Av. charge/head = $\frac{\$2406.25}{115}$
 $= \$20.95$ ✓

QUESTION 7

a) $P(\text{hit by Missile}) = P(M) = 0.85$ $P(\tilde{M}) = 0.15$

$P(\text{hit by gun}) = P(G) = 0.2$ $P(\tilde{G}) = 0.8$

i) $P(\text{plane missed by both}) = 0.15 \times 0.8 \checkmark$
 $= 0.12 \checkmark$



$P(\text{at least hit}) = 1 - P(\tilde{M}\tilde{G}) \checkmark$
 $= 1 - (0.12)^2 \checkmark$
 $= 0.9856 \checkmark$

b) i) $\int_0^4 3^x dx \doteq \frac{h}{2} [f(x_1) + f(x_3) + 2f(x_2)]$
 $= \frac{2}{2} [f(0) + f(4) + 2f(2)]$ $\left\{ \begin{array}{l} h = \frac{b-a}{2} \\ = \frac{4-0}{2} \\ = 2 \end{array} \right.$
 $= [3^0 + 3^4 + 2 \times 3^2]$
 $= 100 \checkmark$

ii) Shaded region $\doteq$ $\overset{\text{area}}{\text{rectangle}} - 100$
 $= 81 \times 4 - 100 \checkmark$
 shaded area $\doteq 224 \text{ u}^2 \checkmark$

Q7

c) i) $y = x^2 - 2x - 3$ (1)
 $y = x + 7$ (2)

$\therefore x^2 - 2x - 3 = x + 7$
 $x^2 - 3x - 10 = 0$
 $(x - 5)(x + 2) = 0$

$\therefore x = 5, x = -2 \checkmark$

when $x = 5, y = 12 \Rightarrow (5, 12)$
 when $x = -2, y = 5 \Rightarrow (-2, 5)$

$\therefore A(-2, 5) \quad B(5, 12) \checkmark$

ii) $A = \int_{-2}^5 (x + 7) - (x^2 - 2x - 3) dx$
 $= \int_{-2}^5 -x^2 + 3x + 10 dx$
 $= \left[-\frac{x^3}{3} + \frac{3x^2}{2} + 10x \right]_{-2}^5 \checkmark$
 $= \left(-\frac{125}{3} + \frac{75}{2} + 50 \right) - \left(\frac{8}{3} + 6 - 20 \right)$
 $= 57 \frac{1}{6} \text{ u}^2 \checkmark$

QUESTION 8.

$$a) i) P(\text{breakcode}) = \frac{1}{10} \times \frac{1}{10} \times \frac{1}{10}$$

$$= \frac{1}{1000}$$

$$ii) P(\text{breakcode}) = \frac{1}{10} \times \frac{1}{2} \times \frac{1}{10}$$

$$= \frac{1}{200}$$

$$b) \text{GS: } T_2 = 6 \quad T_2 = 3$$

$$i) \quad \frac{T_6}{T_3} = 3$$

$$T = 3$$

$$ii) T_2 = ar = 6$$

$$a \times 3 = 6$$

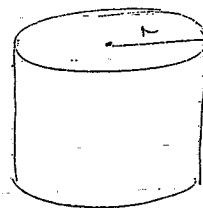
$$a = 2$$

$$iii) S_n = a \frac{(r^n - 1)}{r - 1}$$

$$S_{12} = 2 \frac{(3^{12} - 1)}{3 - 1}$$

$$S_{12} = 531440$$

c)



$$SA = 2\pi r^2 + 2\pi rh$$

$$30 = 2\pi r^2 + 2\pi rh$$

$$15 = \pi r(r + h)$$

$$\frac{15}{\pi} = r + h$$

$$h = \frac{15}{\pi} - r$$

$$ii) V = \pi r^2 h$$

$$= \pi r^2 \left(\frac{15}{\pi} - r \right)$$

must be shown!

$$V = 15r - \pi r^3$$

$$iii) \frac{dV}{dr} = 15 - 3\pi r^2$$

$$\frac{d^2V}{dr^2} = -6\pi r$$

for stationary values $\frac{dV}{dr} = 0$

$$15 - 3\pi r^2 = 0$$

$$3\pi r^2 = 15$$

$$\pi r^2 = 5$$

$$r = \pm \sqrt{\frac{5}{\pi}}$$

$$r = \sqrt{\frac{5}{\pi}} \quad (r > 0) \checkmark$$

$$\text{test } r = \sqrt{\frac{5}{\pi}}, \quad \frac{d^2V}{dr^2} < 0 \Rightarrow \text{max}$$

$$\therefore \text{for max vol, } r = \sqrt{\frac{5}{\pi}}$$

$$= 1.2615 \dots \text{ m}$$

$$r = 12.6 \text{ cm (nearest cm)}$$

QUESTION 9.

9) $\begin{matrix} 20\text{cm} & 1\text{m} & 1.5\text{m} & 1.8\text{m} \\ \text{Now} & 1 & 2 & 3 \end{matrix}$
 1st years growth
 $a = 80\text{cm}$ $r = 0.6$ $n = 3$

$$T_3 = ar^2$$

$$= 80 \times 0.36$$

$$= 176.8 \text{ cm.}$$

OR
$$S_3 = \frac{a(1-r^n)}{1-r}$$

$$= \frac{80\text{cm}(1-0.6^3)}{0.4}$$

$$= 156.8$$

Height = $20 + 156.8$
 $= 176.8 \text{ cm.}$

10) Growth
 $\text{Max} = \frac{a}{1-r}$

$$= \frac{80}{1-0.6}$$

$$= 200$$

$\therefore \text{Max height} = 200 + 20$
 $= 220 \text{ cm.}$

11) $f(x) = x^2 + \frac{1}{x^2}$

i) domain: all reals, $x \neq 0$

ii) $f(-x) = (-x)^2 + \frac{1}{(-x)^2}$

$$= x^2 + \frac{1}{x^2}$$

$\therefore f(x) = f(-x)$
 $\therefore \text{even function}$

✓ must have domain + even fn

iii) as $x \rightarrow 0$ $f(x) \rightarrow \infty$

iv) $f(x) = x^2 + \frac{1}{x^2}$

$$= x^2 + x^{-2}$$

$$f'(x) = 2x - 2x^{-3}$$

$$f''(x) = 2 + 6x^{-4}$$

$$= 2 + \frac{6}{x^4}$$

for stationary values $f'(x) = 0$

$$2x - \frac{2}{x^3} = 0$$

$$2x^4 - 2 = 0$$

$$2(x^4 - 1) = 0$$

$$x^4 = 1$$

$$x = \pm 1$$

if $x = 1$, $y = 2$
 $x = -1$, $y = 2$

$$f''(1) > 0 \Rightarrow \text{min at } (1, 2)$$

$$f''(-1) > 0 \Rightarrow \text{min at } (-1, 2)$$

1 for testing max/min
 1 for y values

v) for concave up $f''(x) > 0$

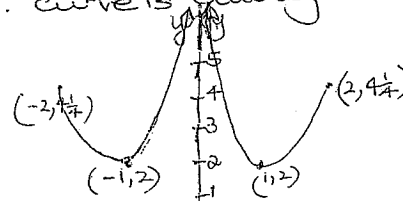
$$2 + \frac{6}{x^4} > 0$$

$$x^4 \text{ is always +ve}$$

$$2 + \text{ve} > 0$$

$\therefore$ curve is always concave up.

vi)



1 for stationary pts + end pts marked
 1 for slope/axes labelled + asymptote.

QUESTION 10

$$P_{2000} = 100\,000$$

$$i) P_{2001} = 100\,000 + 12\% \times 100\,000 - E \\ = 112\,000 - E$$

$$ii) P_{2002} = P_{2001} + 12\% \times P_{2001} - E \\ = P_{2001} (1.12) - E \\ = [112\,000 - E] \cdot 1.12 - E \\ = 125\,440 - 1.12E - E \\ = 125\,440 - 2.12E \\ = 125\,440 - E(1+1.12)$$

$$ii) P_{2006} = 100\,000 \times 1.12^6 - E(1 + 1.12 + \dots + 1.12^5) \\ 140\,000 = 100\,000 \times 1.12^6 - E(S_6) \\ = 100\,000 \times 1.12^6 - E \left[\frac{1(1.12^6 - 1)}{0.12} \right]$$

$$E \left(\frac{1.12^6 - 1}{0.12} \right) = 100\,000 \times 1.12^6 - 140\,000 \\ E = 7070.97$$

$$E = 7071 \text{ (to nearest whole pound)}$$

$$b) i) \frac{d}{dx} (xe^x) = v \cdot \frac{du}{dx} + u \cdot \frac{dv}{dx} \\ = e^x \cdot 1 + x \cdot e^x \\ = e^x + xe^x$$

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$$b) ii) \int \frac{d}{dx} xe^x = \int e^x + xe^x dx$$

$$xe^x = \int e^x dx + \int xe^x dx \\ = e^x + \int xe^x dx$$

$$\therefore \int xe^x dx = (xe^x - e^x) dx$$

$$\text{Now } \int_0^2 xe^x dx = [xe^x - e^x]_0^2 \\ = [(2e^2 - e^2) - (0 - 1)] \\ = e^2 + 1 \text{ (as req'd)}$$

$$V = \pi \int y^2 dx$$

$$= \pi \int_0^2 (e^x + x)^2 dx$$

$$= \pi \int_0^2 (e^{2x} + 2xe^x + x^2) dx$$

$$= \pi \left[\int_0^2 (e^{2x} + x^2) dx + \int_0^2 2xe^x dx \right]$$

$$= \pi \left\{ \left[\frac{e^{2x}}{2} + \frac{x^3}{3} \right]_0^2 + 2(e^2 + 1) \right\}$$

$$= \pi \left[\left(\frac{e^4}{2} + \frac{8}{3} \right) - \left(\frac{1}{2} + 0 \right) + 2(e^2 + 1) \right]$$

$$= \pi \left[\frac{e^4}{2} + 2e^2 + 4\frac{1}{6} \right] u^3$$